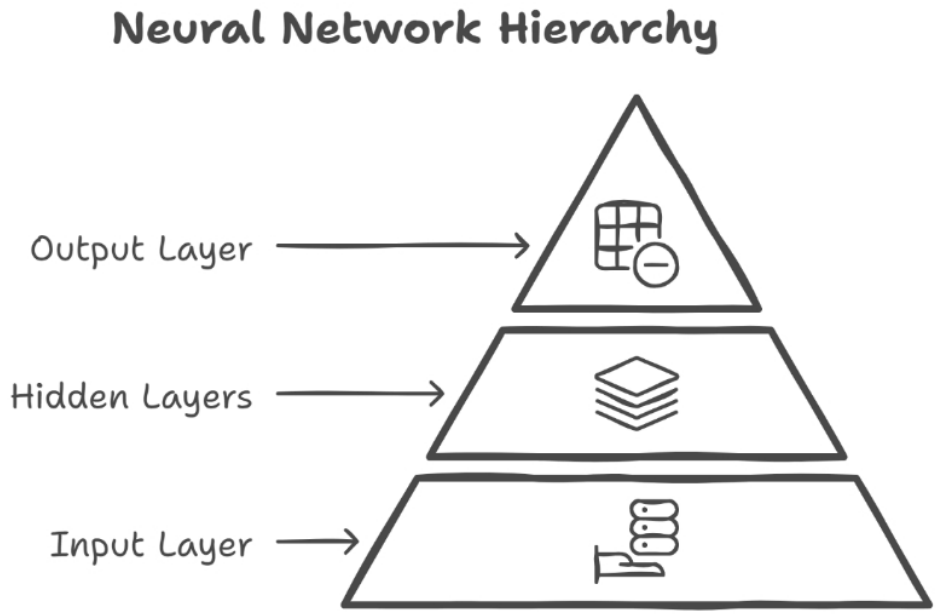
functioning of the human brain. These networks consist of layers of interconnected nodes, referred to as neurons, that work collaboratively to analyze data, identify patterns, and make predictions or classifications. The ability of neural networks to process vast amounts of data and uncover intricate relationships makes them a powerful tool for solving complex problems in diverse fields such as healthcare, finance, and artificial intelligence.



Components of a Neural Network

1. **Input Layer**: The input layer serves as the gateway for raw data to enter the neural network. Each neuron in this layer corresponds to a specific feature of the input data. For example, in an image recognition task, each neuron might represent a pixel value of an image. This layer prepares the data for further processing by subsequent layers.
2. **Hidden Layers**: Hidden layers are the computational backbone of a neural network. These layers consist of neurons that apply mathematical transformations to the input data. Each neuron computes a weighted sum of its inputs, adds a bias term, and passes the result through an activation function. The